

Multi-Size Sheet Metal Can Bracket

Problem

Designing a sheet-metal bracket to hold 12 oz, 16 oz, and 22 oz aluminum cans required balancing stability, manufacturability, cost, and customization. The bracket had to mount to a 1-inch grid, keep each can secure, accommodate a custom-printed resin part, and be manufacturable with a plasma cutter and finger brake. The team also wanted a simple design with few bends, limited material use, and an attractive appearance while remaining competitive with comparable products priced at approximately \$9.97-\$12.99.

Process

After fabricating three candidate designs from cardstock, the team selected Seamus Lee's design as the simplest and most effective. The original concept used two 90-degree bends and no welds, but added support arms required four additional bends and two spot welds. The final bracket used 29.466 square inches of 0.030-inch, 22-gauge A36 steel, cut with a 45-amp plasma cutter at 325 inches per minute and 78 volts. The process then included dross removal, finger-brake bending to 90.884 degrees, spot welding, powder coating, and mounting with #12 screws. Reported time studies ranged from 0:36 internally to 11:20 externally, with an updated estimate of approximately 8 minutes per bracket.

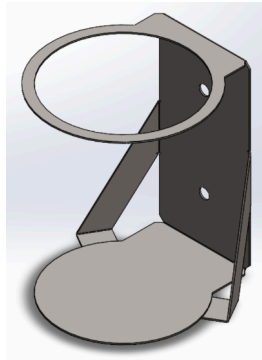


Figure 1: Bracket SolidWorks Model



Figure 2: Bracket Preliminary Cardboard Cutout

Product

The finished bracket met the primary functional requirements by holding 12 oz, 16 oz, and 22 oz cans on a 1-inch grid while providing a stable and durable structure. FEA predicted a minimum factor of safety of 29.09 against yielding under the heaviest expected load. The estimated cost was \$9.641 per unit, including material, fabrication, coating, and a polymer component, supporting a target sale price of \$14.99. The design is practical for a 200-unit low-volume run, but producing 200,000 units would require progressive-die stamping to reduce the estimated cycle time from approximately 8 minutes per part to about 5 seconds.

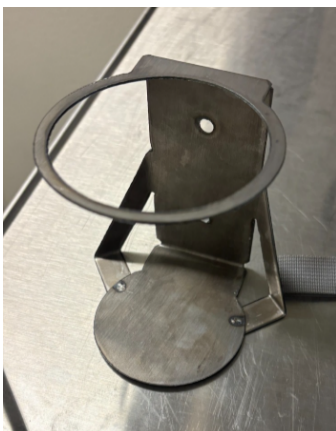


Figure 3: Sheet Metal Bracket



Figure 4: Bracket With Polymer Mount